

Search sheets: how to run the activity



The pages after this one are 17 **search sheets**, built from *The Tale of Peter Rabbit* by Beatrix Potter. Print them one-sided and hand out **one sheet per person** — every sheet is different, and nothing needs cutting out.

If the room is a different size, match it to the sheets rather than the other way round. More people than sheets: print the set twice and hand out both copies. Every token pair is then held by two people, so every count doubles and the proportions — which is all the model is — do not move. It buys some insurance too, because one person away no longer takes a token pair out of the room with them. Fewer people than sheets: give the spares out rather than leaving them on the table, two or three to a person if it comes to that. A sheet nobody holds is a hole in the model.

Each sheet holds 66 **token pairs**. A **token** is one word or one punctuation mark; a token pair is a **next token** paired with the **previous token** that came immediately before it somewhere in the original text.

In **for supper**, the boxed token is the previous token and *supper* is the next token. Each token keeps the same colour wherever it appears, so scan by colour first and read second. With only 11 colours and far more tokens than that, a colour narrows the search rather than settling it — always check the token itself.

Say the colour when you call a token out — “who has *supper*? it’s a teal one” — and the room can find the swatch before reading a word:

black red brown ochre green teal blue navy violet purple wine

Running a round

Seed the board first. Ask someone to read out any token pair from their sheet — both tokens, boxed one first — and write it up. It matters that the opening comes off a sheet rather than out of your head: the room supplies the beginning as well as everything after it. Then, each round:

1. **Call out the last token** of the text so far.
2. Everyone scans their own sheet for token pairs whose previous token matches. **Hands up** — one hand each, and anyone with more than one match just goes with the first they spot.
3. **Pick a raised hand at random** and ask for that token pair’s next token.

4. A scribe writes the token on the board. That token becomes part of the context for the next round. Repeat.

Picking at random matters. If you take whoever shouts first you are sampling the fastest reader, not the text — and the whole point is that the room samples the way the model does.

What to watch for

The hands are the probability distribution. Every token pair was dealt to exactly one person, and the deal spreads each context across as many people as it will go, so a common continuation really does put more hands in the air than a rare one. Nobody has to explain weighted sampling; the room performs it. (Roughly: for a context more common than the class is big, somebody holds two matches but raises one hand, which flattens the busiest contexts.)

Nobody holds the model. No single sheet can continue the text on its own. The model only exists across the whole room — and if somebody is away, their share of it is missing, and some contexts will draw no hands at all.

Quiet rounds. One hand is not a stall — a word with a single continuation is the model working. Nor will you run dry: a word has nothing following it only if it appears nowhere but the very last position in the text, which is at most one word in the whole deal. So no hands at all almost always means an empty seat — two of them, if you printed the set twice — and that is worth saying out loud. Reseed from any token pair and carry on.

1123 tokens, 388 unique · 2.25 bits/token · perplexity 4.8 · branching factor 2.32

Worked example

Start anywhere. Pick any token pair — say this one — and write its previous token *and* its next token on the board.

Round 1. Call out the last token on the board (*supper*). Another sheet has a match, so that person raises a hand and reads out ..

Round 2. Call out the last token on the board (.). Another sheet has a match, so that person raises a hand and reads out *Peter*.

Keep going for as long as you like. The text the room writes will sound like *The Tale of Peter Rabbit* without ever repeating it — that is the model generating, not the corpus being read out.

the token pair that answers

for supper

supper .

Peter

on the board

for supper

for supper .

for supper . Peter

for supper was the if he tired that
, scratch crow to on his he got
the gate there ; , and Mr .
and calling towards Peter very fast carrying peas
some black . Peter such a Peter scuttered
time there net , , who planting out
, went . his lost one , carrying
quietly off a rake twitched as McGregor was
little bunnies . then scritch . frightened ;
noise of thing to but Flopsy cat was
to bed to Peter a dose his water
the floor was busy he might it was
blackbirds . into mischief the shoes wood .
jumped into and milk and climbed flopped down
rabbit took room for never stopped had bread
. Flopsy Kertyschoo ! among the potatoes .
garden . radishes ;

was after

sand bank

Peter was

. rabbit

, leaving

he saw

a very

. rabbit

towards the

the baker's

and got

had done

McGregor caught

too small

alive .

over carefully

very quietly

to frighten

pair of

the least

for some

ate some

, Cotton

time ,

it was

much water

a window

whom should

on the

and jumped

most dreadfully

into the

if it

fortnight !

way to

he tried

old mouse

speaking to

tea ;

jumped up

to Mr

took a

. his

an old

the nice

slipped underneath

. Peter

was very

little rabbit

squeezed under

some parsley

Peter ,

Flopsy ,

there were

, and

along ,

Mr .

garden :

, but

thing he

stone doorstep

and looking

but suddenly

root of

his cousin

no time

find his the gate garden , Peter ;
but don't had lost was tired , and
gold fish old Mrs time . his mother
along a would have ate some a gooseberry
in great the large staring at , and
into the over the of breath bushes .
little Rabbits and then and beyond bunnies ,
for he way straight say that McGregor .
floor of no room . then Flopsy ,
it best an accident lane to to look
very damp Peter , , and some gold
. McGregor . McGregor sitting in still ,
began to on four umbrella , was out
locked , , and the bushes he was
running or . but to squeeze he went
least idea Mr . you may rake and
a wheelbarrow sand on

to him altogether if somewhere in he found
was the Bunny . had not for a
I think some camomile Peter thought running after
pie by lane , McGregor hoeing
, and a sand a pond began to
without speaking very naughty to go
carefully , old Mrs , if and the
, and water in on his by some
Peter asked the window safe at in a
four legs the tool his hands and out
Flopsy , . Mr , Mopsy . McGregor
Mr . over the only shook garden ,
tool shed into Mr it . sorry to
he became bushes . eyes . my dears
turn them way to Also he . Peter
of a caught sight the wood was so
. Also amongst the time he came up

might have he ate began to very still
sobs were jacket . garden , , and
was somewhere Peter , came to some French
to turn French beans dears , way back
Mr . the tip Peter sat sight of
great excitement it was the evening
what he by the cans . in it
running in bread and under each water cans
caught by and Peter the little , and
a cucumber and Cotton , and four little
I am plants . top of , and
tool shed she only climbed upon hung up
wall ; nothing happened Father had
had an across the was staring over .
but his work beans ; it
for a a pie to his of his
he perhaps hidden McGregor , he could

the blackbirds brown bread baker's . bed time

more and shed , to exert Peter had

she sat through the tool shed Peter was

about , came out . McGregor the wood

a scare out young his shoes Cotton tail

very big , and bread and feeling rather

, going McGregor hung Mr . all round

gooseberry net of running was turned the other

hoeing onions sieve , he rushed a time

but now . McGregor for lost by Mrs

Mrs . beans to , but , but

answer . . an ; and and pair

it would soft sand he flopped running as

at some in the under the I am

one of and some buns . jacket and

filled his their names : but round .

to say fast as

the top bed , Mr . . McGregor's
quite close was very sparrows , , scratch
their mother Cotton tail at the calling out
found a very , the stone about cats
or down lived with shed , fright ,
looked behind tears ; he heard one table
and more he had all over pond where
tried to get into : your to hide
. Mr just in ; he loaf of
out of she gave his way jacket with
Once upon but Mr gate , of shoes
to rest . and first he the tool
a can now run Mrs . Peter sneezed
Peter got damp with , said meet but
straight away round the in time and then
more puzzled McGregor ! fast , , but
a time , Mopsy

who jumped became more back towards shed big
doorstep, peeped over out of a blue
outside the Mrs. off the first thing
of Peter was put gate! very well
don't get quite sure young cabbages or looked
jacket and McGregor's garden he had
at last the fields said old been a
going lippity, along cry. the cabbages
. Peter, you in a, but
Cottontail, Mr. wondered what straight walk
the wheelbarrow; he she wondered
but presently and don't scuttered underneath
tried to to the his eyes waving a
and the have been. McGregor now and
one morning knees planting. They Stop thief
him was shoes that, ran Peter did
foot upon puzzled. naughty, he was
to her

the end shed , now my . she
jacket behind . McGregor's rest ; sure that
was safe them , out . , she
but Peter at bed don't go mischief .
to the ; but with their back was
thief ! straight across second little , he
gate , he went good little and went
hidden underneath not answer have got
basket and the garden sick , as fast
shut his , as should he shoes among
brass buttons can . were alive upon the
and ran who was her umbrella she could
his sobs made some , perhaps a loaf
of a him . . McGregor care .
! Mr in the Peter ! going out
close to implored him five currant quite new
McGregor's garden he intended to be

gate . she had can . and trembling
and squeezed with sitting , quite he ran
buttons on done with squeeze underneath
shoes for he came think he not had
thought it them over back to idea which
; and fish , as if up with
was running blackberries for corner , end of
lippity lippity to go , underneath him till
gather blackberries the garden , who
lettuces and large buttons that Peter in the
Peter wriggled ! one his foot were Flopsy
. now after Peter beyond him her head
upon Peter out just but his at him
ran straight spoonful to who flew saw was
to cry away altogether a door . presently
, upsetting friendly sparrows underneath .
well during and knees . I of a
the wood

lippy not he was tail twitched till he
underneath the a straight her the as nothing
she bought , who to wander and there
ran on the second ; and up the
with brass , feeling that Peter buttons ,
of her gate ! , and were four
currant buns hide in in and run along
who were evening . away without taken at
was on the potatoes not very large pea
upon a after a clothes . but he
faster , and five . after him .
cousin , his back supper . his
out , and was wander about window ,
stopped running , waving he slipped
blackberries : Peter began . he to the
! I back to got caught went back
nice soft peas and . he

away to to pop Mopsy , ran after
her family out , were good bought a
each . but Peter in that window was
he began sat down himself . . after
went faster Peter never currant bushes , he
tail had the gate and made was locked
him to that he up for a fortnight
! Mr r ritch he was to her
of him fields or his mother run into
scratch , with fright . Peter got home
and he ; but Benjamin Bunny
! McGregor after him ! McGregor , which
had heard down the as he , whom
underneath the , but did not look for
morning , intended to and started cabbages ,
upon the not unfortunately turned towards
the corner flower pot work . fir tree

Rabbits , Mopsy , and Cottontail he came
then he the rabbit put him mother was
with his underneath the and beans go ,
, scritch him , . it . the
not very , and , looking other shoe
scratch , his jacket tree . went to
leaving his up and sat very bank ,
fir tree looking all himself up ! Peter
Peter gave her mouth he lost down upon
of brown upon a got down in a
pop upon that I but round cabbages ,
tail , to find scr r door in
; but the big was a to a
may go out over after Peter were overheard
heard the shook her and shut had such
. presently a hoe am going . a
your Father , who

head at new . Peter was got away
, but and she . he from his
heard about had not and then ! first
a white ritch , of a and tried
Mr . he began breath and underneath a
down very him at went back camomile tea
, and . the frighten the his clothes
looking under cooking ; put his mouth that
am sorry tree . . now asked her
accident there ; he . he , and
with a but it to him big fir
the gate the first in , , very
mother in then old names were go into
that she presently Peter a wall flew to
not care was no milk and forgotten the
McGregor was Mopsy , after losing he jumped
her ; white cat

be taken not the rabbit to blue jacket
behind some mother put the root but Peter
could go presently , sneezed Kertyschoo
some radishes went down excitement , beautiful thing
They lived wriggled out he went , Mopsy
to gather table spoonful his jacket dose of
he had . Mr ! Peter rushed all
the lane in a . but and he
the way her tail had so go .
. it , he during the ; he
and blackberries wheelbarrow and
a sieve , he . Mr that Peter
tip of so much was quite a large
put in little jacket him in overheard by
which he Peter . down the then the
McGregor was , Stop of it Mr .
it were fat little and shed big tears
frame ,

pea in some lettuces the lane to the
, for go away but she he meet
, so he had cucumber frame
then he ; and the garden gave a
was Mr rather sick lost , went through
Peter . could not suddenly , where Mr
onions . Mr . the noise it to
down to busy cooking in no little Benjamin
pot . losing them . and and implored
rushed into behind him a flower McGregor filled
was too presently , him in small for
a beautiful and her which way big fir
. McGregor dreadfully frightened into a
so that wheelbarrow ; that can he was
, and shoe amongst . McGregor exert himself
had forgotten . she of the black currant
, and and peeped wood to rabbit one

gave himself so tired cats from into a
and rushed there was . he behind him
lost in tired of little jacket scare crow
hoe scr had not happened , Peter ,
go into walk behind best to some friendly
him to . he time . McGregor came
, and hands and he ate and went
if it upsetting three ; she , he
the gate started running . Peter it had
the way a basket . Peter was most
unfortunately run the tool mouse was
Mr . , little but he parsley .
for Mr three plants a fat legs and
and their wood outside , quite then ,
in her jumped out to put home to
rabbit hole he had was not last in
family in hole and trembling with